

Homework 8. ARIMA

You can get up to 10 points for this homework.

Question 1. Fit an ARIMA model for the series given in the Table 1, representing data from the US Census Bureau on monthly domestic automobile manufacturing shipments (in millions of dollars). Make all necessary steps, plots, predictions and comparison. Use Python and submit one file incorporating all your answers, named as "H8-YourFirstName-YourLastName.ipynb".

- ☐ A Create a .csv file out of the given data and put it online (you will use a link to it in your program).
- ☐ B Plot the data.
- ☐ C Use time series decomposition to identify the presence of a periodic pattern. Why does a seasonal behavior occur?
- ☐ D Find the order of integration.
- ☐ E Use the `optimize_ARIMA` function and select the model with the lowest AIC.
- ☐ F Perform a residual analysis.
- ☐ G Develop an ARIMA model for forecasting shipments.
- ☐ H Compare the performance of the ARIMA model against a baseline model of your choice. Use the MAPE.

TABLE 1. US Automobile Manufacturing Shipments (Dollar in Millions)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1992	5618	7205	7446	7420	8076	7692	4395	7045	7058	8219	7432	6427
1993	7091	8339	8231	7772	8012	8015	3831	6246	7399	8546	8186	7267
1994	7986	9482	9215	9092	9346	9653	4748	8832	9194	9583	9573	8560
1995	8541	10,048	9796	8781	8851	8877	4207	7898	8999	9167	8333	7236
1996	7379	8852	7817	8688	8993	8521	4900	7837	8895	8739	8213	6,425
1997	7698	8898	8228	8121	7804	8120	5151	7943	8179	9354	8490	7,380
1998	7248	9013	9280	8622	8706	7312	3874	8162	9006	9658	8396	7431
1999	7594	9144	9429	8550	8635	8738	4657	8797	8895	9280	8607	7477
2000	8231	9117	9817	8467	8919	9536	4437	7941	9267	9133	7720	6175
2001	6485	7142	8541	6986	8431	7999	4411	8182	6827	8372	7312	6358
2002	7227	7239	7541	7886	8382	7726	4982	8089	7465	8565	7149	5876
2003	7383	7308	7923	6959	7638	7374	4903	6937	8036	8520	6680	6575
2004	6670	7778	9103	7323	7306	7947	4212	7709	7875	7453	6837	6634
2005	6520	7302	7692	6966	7403	7906	4680	8289	8402	8472	7525	7209
2006	7454	8009	9829	7571	9143	9247	4424	8655	7621	8126	7278	6872
2007	6788	6897	8121	6468	7424	7482	4945	8010	6969	7962	7250	6412
2008	7328	7862	7229	6818	7092	7376	5156	7302	7386	7352	5673	4930
2009	3869	3995	4221	4039	3519	3994	3277	4635	4911	5554	5106	5287
2010	5373	5021	5969	5716	5788	6765	5397	7262	7422	6957	6389	6171
2011	5974	6324	8197	6448	6752	7521	5739	7761	7508	8362	7288	7253
2012	7308	8290	8919	8659	9594	9678	7141	9437	8886	10,081	9,898	9085
2013	9789	10,591	11,387	10,666	10,920	10,992	8076	11,166	10,989	12,195	10,996	8365
2014	8683	9705	9577	8356	9686	—	—	—	—	—	—	—